# Stage 8 Maths — Practice Test 2

Paper 1: 60 minutes · no calculator · Paper 2: 60 minutes · calculator allowed · Total 100 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 100

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

## Paper 1 — 60 minutes — 50 marks

**1** **[1]**

(a) Work out $20 - 4 \times 3^2 \div 6$

Working

Answer

(b) Brackets are added to $5 + 3 \times 4 - 2$ so that the answer is 30. Which one is correct? **[1]**

- ☐ A) $(5 + 3) \times 4 - 2$
- ☐ B) $5 + (3 \times 4) - 2$
- ☐ C) $5 + 3 \times (4 - 2)$
- ☐ D) $(5 + 3) \times (4 - 2)$

**2** **[1]**

(a) Work out $(-84) \div (-4) - 3 \times (-5)$

Working

Answer

(b) Ali says: “When you multiply three negative integers together, the answer is always negative.” **[1]**
Is Ali correct? Explain how you know.

**3** The Hassan family drive from Cairo to Alexandria for their summer holiday. The road distance is about 200 km. **[1]**

Use 1 km ≈ 5/8 of a mile to find the distance in miles.

Working

Answer ____ miles

**4** On holiday the Hassan family drink 2 3/4 litres of bottled water each day. **[2]**

(a) How many litres do they drink in 6 days? Give your answer as a mixed number.

Working

Answer ____ litres

(b) They pour a 9-litre bottle of water into cups. Each cup holds 3/4 of a litre. How many cups can they fill? **[1]**

Working

Answer ____

(c) On Monday the family walk 4 1/3 km along the Corniche. On Tuesday they walk 1 3/4 km. How much further do they walk on Monday? Give your answer as a mixed number in its simplest form. **[2]**

**5** A hotel room in Alexandria cost E£800 a night last summer. This summer the same room costs E£920 a night. **[1]**

(a) Work out the absolute change in the price.

Working

Answer E£

(b) Work out the percentage increase in the price. **[2]**

Working

Answer %

**6** **[1]**

(a) $5^7 \times 5^2 \div 5^4 = 5^n$. Find the value of n.

Working

Answer

(b) Write down the value of $\sqrt[3]{(-64)}$. **[1]**

Working

Answer

(c) Work out $(-5)^2 - (-2)^3$ **[1]**

Working

Answer

**7** **[1]**

Work out 0.72 ÷ 0.01

Working

Answer

**8** **[1]**

(a) Which fraction is equal to a recurring decimal?

- ☐ A) 3/8
- ☐ B) 7/20
- ☐ C) 5/12
- ☐ D) 9/25

(b) Zara says: “7/30 is a terminating decimal, because 30 is a multiple of 10.” **[1]**
Is Zara correct? Explain your answer.

(c) Which list is in order, starting with the smallest? **[1]**

- ☐ A) −0.6, −5/8, −0.65, −2/3
- ☐ B) −0.65, −2/3, −0.6, −5/8
- ☐ C) −5/8, −0.6, −2/3, −0.65
- ☐ D) −2/3, −0.65, −5/8, −0.6

**9** **[2]**

Work out 8.64 ÷ 0.4

Working

Answer

**10** **[1]**

Show how to work out $3.7 \times 46 + 3.7 \times 54$ without a calculator. Write down the answer.

**11** **[1]**

(a) Which one of these is a formula?

- ☐ A) $4x + 3$
- ☐ B) $4x + 3 = 19$
- ☐ C) $x > 4$
- ☐ D) $P = 4s$, where P is the perimeter of a square of side s

(b) Simplify $3a^2 + 5a - a^2 + 2a$ **[1]**

(c) Expand $2x(x^2 - 3)$ **[1]**

(d) Factorise completely $12y^2 + 18y$ **[2]**

**12** **[2]**

(a) Solve $5x - 7 = 2x + 11$

(b) Solve $(x + 4)/3 = 5$ **[1]**

Working

Answer

**[1]**

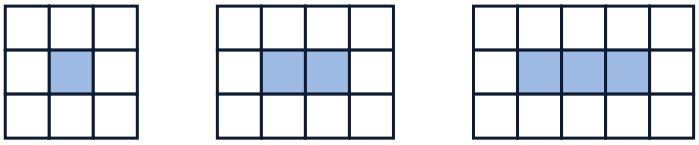



(c) The diagram shows an inequality on a number line.
Which inequality is shown?

- ☐ A) $-2 \leq x < 3$
- ☐ B) $-2 < x \leq 3$
- ☐ C) $-2 \leq x \leq 3$
- ☐ D) $-2 < x < 3$

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**13** Rashid makes patterns with square tiles. Each pattern has a row of blue tiles with white tiles all round it. **[1]**



Pattern 1 Pattern 2 Pattern 3

(a) How many white tiles are in Pattern 5?

Working

Answer

(b) Write an expression for the number of white tiles in Pattern n. **[2]**

(c) Rashid says one of his patterns has exactly 75 white tiles. Explain why he cannot be correct. **[1]**

**14** **[1]**

(a) Here is a function machine:
input → × 3 → − 4 → output
The output is 17. Work out the input.

Working

Answer

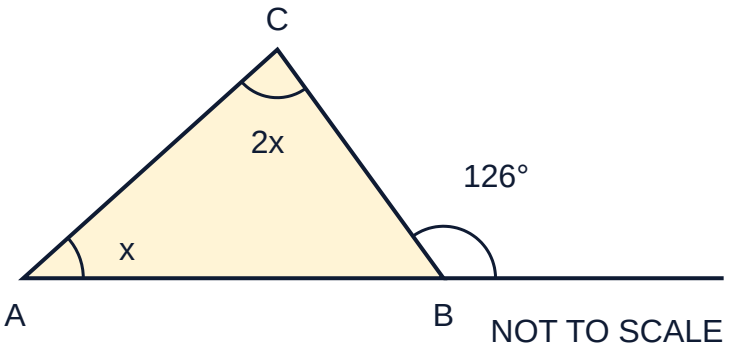
(b) Which line is parallel to y = 3x − 2? **[1]**

- ☐ A) y = 2x − 3
- ☐ B) y = 3x + 5
- ☐ C) y = −3x − 2
- ☐ D) y = x + 3

**15** Omar is drawing a tiling pattern. He says: “A rhombus is a special kind of square.” **[1]**

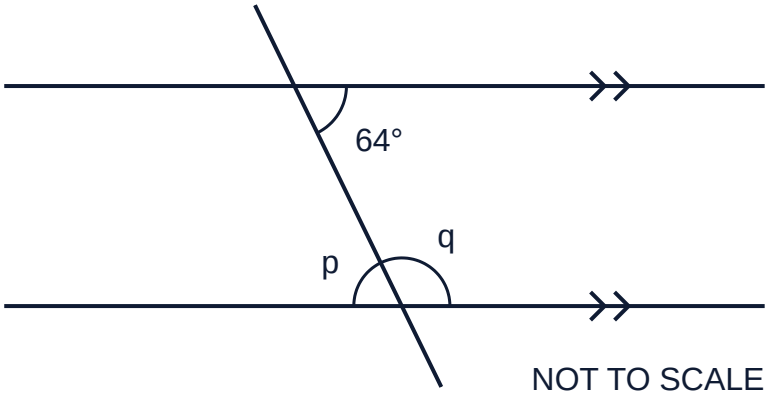
Explain why Omar is wrong.

**16** In triangle ABC, AB is extended. Angle A = x and angle C = 2x. The exterior angle at B is 126°. **[2]**




Work out the value of x.

**17** The diagram shows two parallel lines crossed by a straight line. **[1]**




(a) Work out the size of angle q.

Working

Answer ° 

(b) Angle p = 64°. Give a geometrical reason. **[1]**

**18** Use a ruler and compasses only. Do not rub out your construction arcs. **[2]**

Construct triangle ABC with AB = 7 cm, AC = 5 cm and BC = 6 cm.

**19** **[1]**

(a) A boat sails from the harbour in Alexandria on a bearing of 065° to a fishing area.
Work out the bearing of the harbour from the fishing area.

Working

Answer ° 

(b) On a plan of a garden, a straight path goes from P(−3, 4) to Q(5, −2). A bench is at the midpoint of PQ. **[1]**
Work out the coordinates of the bench.

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**20** The school canteen sells a meal deal. Students choose one main and one drink. **[1]**
Mains: rice, pasta, sandwich
Drinks: water, juice, milk, lemonade

(a) List all the possible meal deals, or show them in a table.

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(b) Laila chooses a main and a drink at random. Find the probability that she chooses pasta and juice. **[1]**

Working

Answer

(c) The probability that the canteen sells out of sandwiches on a school day is 0.15. **[1]**
Find the probability that the canteen does not sell out of sandwiches on a school day.

Working

Answer

## Paper 2 — 60 minutes — 50 marks

**1** Uncle Sami visits from London for the Hassan family's holiday in Alexandria. On the day he changes money, £1 = E£61.40. **[1]**

(a) Uncle Sami changes £250 into Egyptian pounds. How many Egyptian pounds does he get?

Working

Answer ____ E£

(b) A boat trip costs E£450 per person. Uncle Sami pays for 5 people. Work out the total cost in pounds (£). Give your answer to the nearest penny. **[2]**

Working

Answer ____ £

(c) A beach hotel welcomed 38 549 guests last year. **[1]**
Round 38 549 to 3 significant figures.

Working

Answer ____

**2** The price of a train ticket from Cairo to Alexandria goes up from E£85 to E£102. **[2]**

Work out the percentage increase.

Working

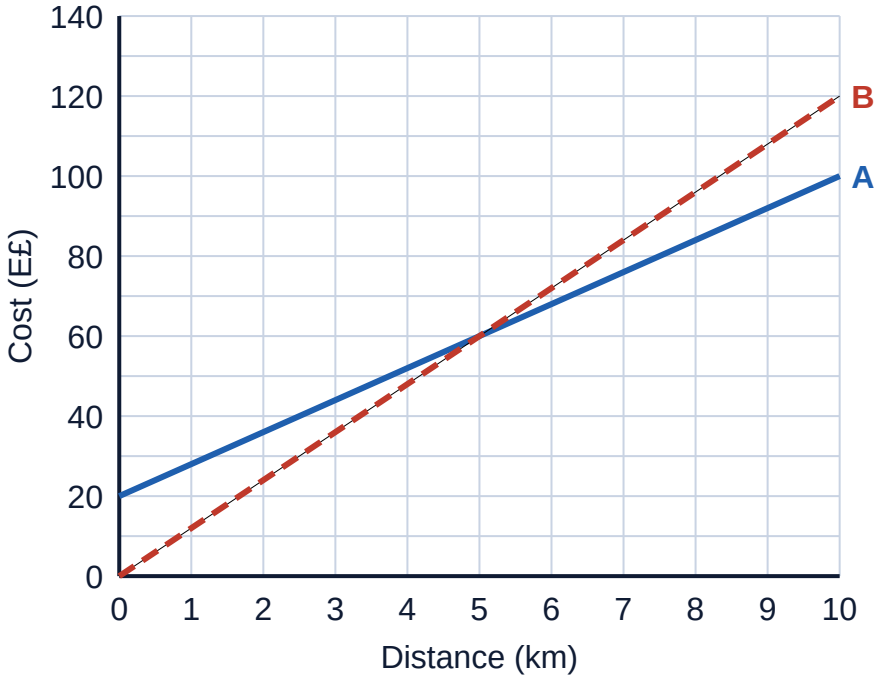
Answer ____ %

**3** Two taxi companies in Alexandria charge like this: **[1]**

Company A: E£20 plus E£8 for each kilometre

Company B: E£12 for each kilometre

The graph shows the cost of a journey for each company.




(a) Use the graph. How far can you travel with Company A for E£100?

Working

Answer ____ km

(b) For what distance do both companies charge the same amount? **[1]**

Working

Answer ____ km

(c) Mona takes a 9 km journey. Which company is cheaper, and by how much? Show your working. **[2]**

(d) For Company A, the cost C (E£) for d km is C = 20 + 8d. **[1]**
Make d the subject of the formula C = 20 + 8d.

(e) The Hassan family also compare phone plans. Plan Z costs a fixed E£30 each month plus E£5 for each GB of data used. **[1]**
y is the monthly cost in E£ and x is the number of GB used. Write an equation for y in terms of x, in the form y = mx + c.

**4** A school cycling club takes part in a race. Karim's bicycle wheel has a diameter of 68 cm. **[2]**

(a) Work out the circumference of the wheel. Give your answer correct to 1 decimal place.

Working

Answer ____ cm

(b) One stage of the race is 5 km long. How many complete turns does the wheel make in this stage? **[2]**

(c) On a track, Rider P completes a lap every 6 minutes and Rider Q completes a lap every 8 minutes. They start together on the start line. **[1]**
After how many minutes are they next on the start line together?

Working

Answer ____ minutes

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**5** The race times, in minutes, of the 15 riders in the Red team are shown in the stem-and-leaf diagram. **[1]**

| Stem | Leaf |
|---|---|
| 3 | 8 |
| 4 | 1 3 7 |
| 5 | 0 2 2 5 8 |
| 6 | 0 1 4 9 |
| 7 | 2 5 |

Key: a stem of 4 with a leaf of 1 means 41 minutes

(a) Find the median time of the Red team.

Working

Answer ____ minutes

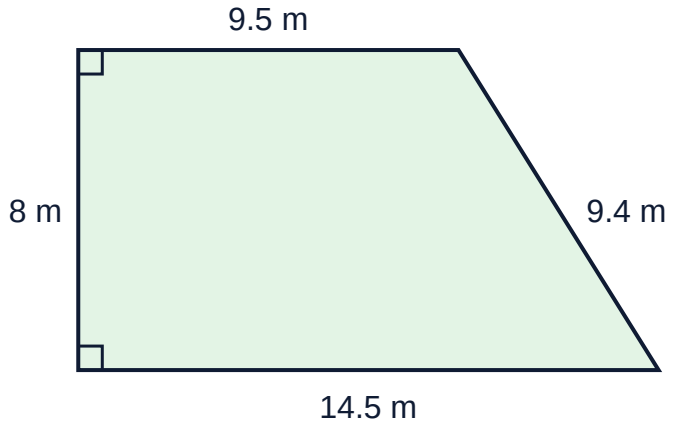
(b) Find the range of the Red team's times. **[1]**

Working

Answer ____ minutes

(c) The Blue team's times have a median of 58 minutes and a range of 21 minutes. **[2]**
Compare the times of the two teams. Write two comparisons.

**6** Nour is designing a garden. The garden is a trapezium. **[2]**




(a) Work out the area of the garden.

Working

Answer ____ $m^2$

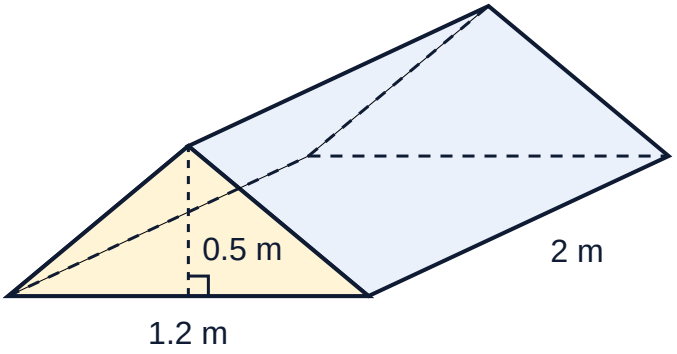
(b) Nour puts fencing all the way round the garden, except for a gate 3 m wide. Fencing is sold in panels 1.8 m long. **[2]**
How many panels does Nour need to buy?

(c) Inside the garden, a flowerbed is a parallelogram with base 3.6 m and perpendicular height 2.5 m. **[1]**
Work out the area of the flowerbed.

Working

Answer ____ $m^2$

**[2]**




(d) A rockery is a triangular prism made of soil. The triangle has base 1.2 m and height 0.5 m. The prism is 2 m long.
Work out the volume of soil in the rockery.

Working

Answer ____ $m^3$

(e) Nour makes a planter box with no lid. It is a cuboid 80 cm long, 40 cm wide and 30 cm high, made of wood on the base and the four sides. **[2]**
Work out the area of wood needed.

Working

Answer ____ $cm^2$

(f) A rectangular vegetable patch is $(2x + 3)$ m long and $(x - 1)$ m wide. Its perimeter is 25 m. **[2]**
Write an equation and solve it to find x.

(g) A garden lamp is a 3D shape with 12 vertices and 18 edges. **[1]**
Use Euler's formula to work out the number of faces.

Working

Answer ____ faces

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**7** A tile designer works with square tiles. A rectangular wall panel is 84 cm by 120 cm. It is covered exactly with identical square tiles, with no cutting. The tiles are as large as possible. **[1]**

(a) What is the side length of each tile?

Working

Answer ____ cm

(b) How many tiles cover the panel? **[1]**

Working

Answer ____ tiles

(c) One design uses regular octagons. **[1]**
Write down the order of rotational symmetry of a regular octagon.

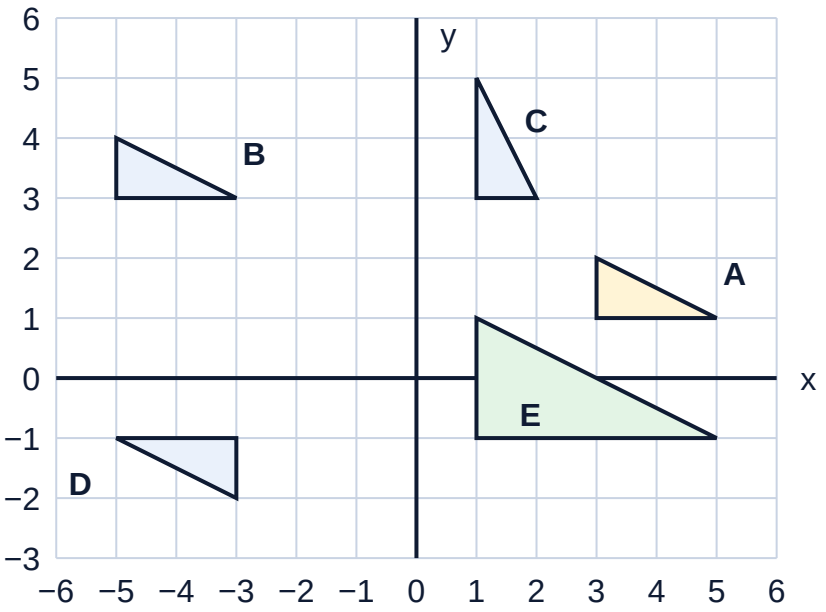
Working

Answer ____

(d) Tiles come in boxes of 24. Hana buys n boxes and uses all the tiles except 7. **[1]**
Write an expression, in terms of n, for the number of tiles Hana uses.

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**8** A tile designer draws triangle A and four images of it, B, C, D and E, on a coordinate grid. **[1]**




(a) Triangle A is translated to triangle B. Write down the vector of this translation, as (x, y).

(b) Describe the single transformation that maps triangle A onto triangle C. **[1]**

- ☐ A) Rotation 90° anticlockwise about (0, 0)
- ☐ B) Reflection in the y-axis
- ☐ C) Reflection in the line y = x
- ☐ D) Translation by vector (−2, 2)

(c) Describe the single transformation that maps triangle A onto triangle D. **[1]**

- ☐ A) Rotation 180° about (0, 0)
- ☐ B) Reflection in the x-axis
- ☐ C) Reflection in the y-axis
- ☐ D) Translation by vector (−8, −2)

(d) Triangle E is an enlargement of triangle A. Write down the scale factor of the enlargement. **[1]**

Working

Answer

(e) Find the coordinates of the centre of the enlargement that maps A onto E. **[1]**

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**9** The canteen manager surveys all 120 students in Year 8 about lunch. Some results are in the two-way table. **[1]**

| | **Hot meal** | **Packed lunch** | **Total** |
|---|---|---|---|
| Boys | 38 | | 60 |
| Girls | | 31 | |
| Total | 67 | | 120 |

(a) How many girls have a hot meal?

Working

Answer

(b) One of the 120 students is chosen at random. Find the probability that the student has a packed lunch. **[1]**

Working

Answer

(c) The manager plans some more survey questions. **[1]**
Which question gives continuous data?

- ☐ A) What is your favourite canteen meal?
- ☐ B) How long, in minutes, did you wait in the queue today?
- ☐ C) How many times did you buy a drink this week?
- ☐ D) Which year group are you in?

(d) To find out what the whole school thinks about the canteen, the manager asks the first 30 students in the queue on Monday. **[1]**
Give one reason why this sample may not represent the whole school.

(e) In a survey of 160 students, 52 chose rice as their favourite meal. The results are shown in a pie chart. **[1]**

Work out the angle of the sector for rice.

Working

Answer ° 

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**10** Laila makes a spinner with 4 equal sections numbered 1, 2, 3 and 4. She spins it 200 times. It lands on 4 a total of 62 times. **[1]**

(a) Write down the experimental probability of the spinner landing on 4.

Working

Answer

(b) If the spinner is fair, how many times would you expect it to land on 4 in 200 spins? **[1]**

Working

Answer

(c) Laila wants a more reliable estimate of the probability of landing on 4. What should she do? **[1]**